2024 Bi H1 Q1

Section: DNA and the Genome

Topic: Structure of DNA

Question Summary:

In a DNA molecule, phosphate groups are found at which end and joined to which part of the nucleotide?

Worked Solution:

- Each DNA nucleotide has a deoxyribose sugar with numbered carbons 1' to 5'.

- The phosphate group is attached to the 5' carbon of deoxyribose and forms the phosphodiester backbone.

- The 3' end exposes a hydroxyl group on the sugar, not a phosphate.

Therefore, phosphate groups are at the 5' ends and are joined to deoxyribose sugar.

Final Answer: **D** (5' ends and joined to deoxyribose sugar).

Revision Tips:

- Remember: 5' end -> phosphate; 3' end -> hydroxyl (OH).

- Bases attach to the 1' carbon of deoxyribose.

- The sugar-phosphate backbone runs antiparallel: one strand 5' -> 3', the other 3' -> 5'.